

Autonomous Aerial System for Precision Crop Monitoring

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Chapter 1

Introduction

In the realm of modern agriculture, crop health monitoring and precision agriculture have emerged as crucial components for optimizing yields, reducing resource usage, and ensuring sustainable farming practices. Traditional methods of monitoring crop health and managing agricultural operations often rely on manual inspections and subjective assessments, leading to inefficiencies, inaccuracies, and increased operational costs.

To address these challenges and usher in a new era of smart agriculture, we propose the development of an autonomous aerial system for precision crop monitoring. The main objective is to deploy an innovative drone system that autonomously scans farmland, captures high-resolution images and videos using a gimbal-supported camera, and processes this data to generate a comprehensive report on the health of the crops.

Our approach involves leveraging advancements in drone technology, image processing algorithms, and data analytics to revolutionize crop monitoring and agricultural management practices.

The project involves the custom design of a 5kg drone, meticulously considering our specific requirements, environmental conditions, and other relevant factors. We aimed to design a drone that could withstand varying weather conditions, efficiently cover large agricultural areas, and seamlessly integrate with our image processing and data analysis systems. Additionally, to gain valuable hardware implementation experience and validate our design choices, we began by assembling a smaller 2kg drone and testing its capabilities, including live video streaming.

In the following sections of this report, we will discuss in detail the design and implementation of our autonomous aerial system. Additionally, we will explore the potential applications of our system, the challenges encountered during the development process, and the strategies employed to overcome them.

Chapter 2

Project Objectives

The project "Autonomous Aerial System for Precision Crop Monitoring" aims to achieve the following objectives:

1) Develop a Custom-Designed 5kg Autonomous Drone: This objective focuses on the design and construction of a 5kg drone specifically tailored for our agricultural monitoring needs. The design considerations will include factors like payload capacity, flight time, maneuverability, and stability in varying weather conditions.

2) Integrate High-Resolution Camera with Gimbal Stabilization: This objective involves integrating a high-resolution camera with a gimbal stabilization system onto the custom-designed drone. The gimbal will ensure stable and focused image capture during flight, even in turbulent conditions.

3) Develop Autonomous Flight Path Planning: This objective focuses on developing software algorithms for autonomous flight path planning for the drone. The software will consider factors like field size, shape, and desired image overlap to generate efficient flight paths for comprehensive data collection.

4) Implement Image Processing Algorithms for Crop Health Assessment: This objective involves developing and integrating image processing algorithms at the ground station. These algorithms will analyze captured images from the drone, identifying features such as color variations, leaf size and shape, and potential disease or pest damage.

5) Generate Comprehensive Crop Health Reports: This objective focuses on developing software functionalities to analyze the results from the image processing algorithms. This analysis will be used to generate detailed reports on the health of the crops across the entire field, highlighting potential problem areas.

6) Evaluate System Performance and Accuracy: This objective involves testing and evaluating the overall performance and accuracy of the developed system. This includes testing the drone's flight stability, image capture

quality, and the accuracy of the generated crop health reports compared to ground-truth data.

7) Demonstrate Feasibility of Precision Agriculture Applications:

This objective involves utilizing the developed system in a simulated or real-world agricultural setting. The project will demonstrate how the generated crop health reports can be used for targeted resource applications, such as fertilizer or pesticide use, promoting precision agriculture practices.

8) Analyze Cost-Effectiveness and Scalability Potential: This objective focuses on analyzing the cost-effectiveness and scalability potential of the developed system. This will involve evaluating the hardware and software costs, as well as assessing the system's feasibility for implementation across various farm sizes and agricultural environments.

By achieving these objectives, this project aims to present a viable and efficient solution for crop health monitoring using an intelligent aerial system. This technology has the potential to revolutionize agricultural practices, promote sustainable resource management, and contribute to improved crop yields and overall agricultural productivity.

Chapter 3

System Design

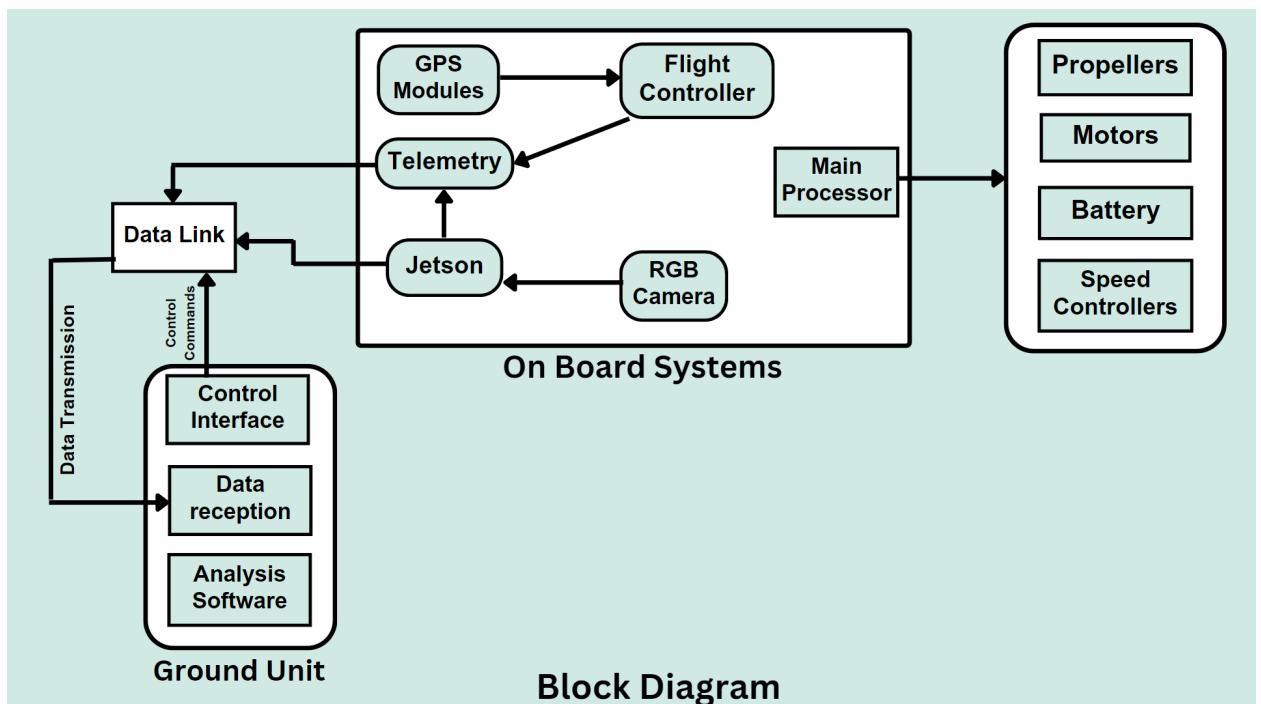


Figure 3.1: Block Diagram of the System

Drone Platform Overview

The Drone Platform serves as the main operational unit in the field. It houses several key components essential for its autonomous and remote-controlled operations in agricultural monitoring and analysis.

On-Board Systems

- **Flight Controller:** Serves as the central processing unit for the drone, managing flight patterns, stabilization, and responses to navigation com-

mands based on data received from the GPS Module and the IMU (Inertial Measurement Unit).

- **GPS Module:** Provides real-time geographical positioning data to the Flight Controller, enabling precise location tracking necessary for targeted agricultural surveying.
- **IMU (Inertial Measurement Unit):** Feeds motion and orientation data to the Flight Controller, enhancing flight stability and accuracy.
- **Camera:** Captures high-resolution images of the agricultural land. These images are crucial for monitoring crop health and other land attributes.
- **Jetson (On-Board Computer):** Processes the images captured by the Camera in real-time, utilizing advanced algorithms for image analysis and data extraction relevant to agricultural assessments.
- **Telemetry System:** Facilitates bi-directional communication between the drone and the Ground Station, transmitting operational data and receiving control commands.

Data Link Connectivity

Connects the on-board Telemetry System and Jetson to the Ground Station. It is responsible for the real-time transmission of processed data and telemetry information to the Ground Station and receiving control commands from it.

Ground Station Block Diagram Explanation

Ground Station Overview

The Ground Station acts as the command and control center for the Drone Platform, equipped with systems for data reception, control interface, and data analysis.

Components

- **Control Interface:** Allows operators to send control commands to the Drone via the Telemetry System. It provides a user interface for live interaction with the Drone's flight operations and camera control.

- **Data Reception:** Receives processed data and raw data streams from the Drone via the Data Link. This includes both image data processed by the Jetson and telemetry data from the drone's operations.
- **Analysis Software:** Analyzes received data for further agricultural insights. The software processes the data to produce actionable information, such as crop health indicators, which can inform farming decisions.

Communication and Data Flow

- **From Drone to Ground Station:**
 - The Jetson processes image data and sends the processed information to the Ground Station via the Data Link, facilitating real-time decision-making.
 - The Telemetry System sends operational data to the Ground Station, ensuring that the drone's status and activities are monitored continuously.
- **From Ground Station to Drone:**
 - Control commands are transmitted from the Control Interface of the Ground Station to the Drone's Telemetry System via the Data Link, enabling remote piloting and real-time adjustments to drone operations and camera settings.

Drone Design

A critical factor in achieving successful drone flight is ensuring sufficient lift to overcome the drone's weight. This is quantified by the thrust-to-weight ratio (TWR). A TWR greater than 1 indicates that the drone's total thrust exceeds its weight, allowing it to ascend and maneuver. The higher the TWR, the greater the drone's acceleration, payload capacity, and ability to handle wind gusts.

In this project, we have chosen a TWR of 2:1. This means the total thrust generated by the drone's motors must be double its weight for optimal performance. This approach provides several advantages:

Efficient Takeoff and Climb:

A TWR of 2:1 ensures the drone can efficiently ascend with adequate reserve power for maneuvering and climbing. **Stable Flight:** This ratio allows for better control and stability during flight, particularly in windy conditions. **Payload Capacity:** The additional thrust provides headroom for carrying payloads like cameras or sensors without compromising flight performance. To achieve this desired TWR, we employed the following calculation process:

Determine Drone Weight:

The total weight of the drone, including all components (frame, motors, batteries, electronics, etc.), is measured in kilograms (kg). In our case, the total weight is 5kg. **Calculate Required Thrust:** Multiply the drone's weight by the desired TWR (2:1). This calculation yields the total thrust required for the drone to achieve the targeted performance, which in this case is $5\text{kg} * 2 = 10\text{kg}$. By following these steps and selecting components that collectively generate the required 10kg of thrust, we ensure the drone possesses the necessary power for efficient flight and payload capacity for our crop health monitoring application.

Chapter 4

Mathematical Model

4.1 Coordinate System and Notations

- **Body Frame (B):**

- This coordinate system is fixed to the drone. It rotates with the drone and moves as the drone moves.
- The forces and torques generated by the drone's motors are expressed in this frame.
- Example: The thrust generated by the motors acts along the z-axis of the body frame.

- **Inertial Frame (I):**

- This coordinate system is fixed to the Earth.
- Used for analyzing the drone's position, velocity, and trajectory in the real world.
- Example: The effect of gravity is defined in the inertial frame, pointing down in the Earth's z-direction.

Notations

- **Position in the inertial frame:** $\mathbf{p} = [x, y, z]^T$
- **Velocity in the inertial frame:** $\mathbf{v} = \frac{d\mathbf{p}}{dt} = [v_x, v_y, v_z]^T$
- **Acceleration in the inertial frame:** $\mathbf{a} = \frac{d\mathbf{v}}{dt}$
- **Angular velocity in the body frame:** $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$
- **Orientation:** Represented by Euler angles ϕ (roll), θ (pitch), and ψ (yaw).

- **Forces in the body frame:** $\mathbf{F} = [F_x, F_y, F_z]^T$
- **Torques in the body frame:** $\boldsymbol{\tau} = [\tau_\phi, \tau_\theta, \tau_\psi]^T$

4.2 Translational Dynamics

Force Balance

Newton's second law governs the translational motion:

$$m\mathbf{a} = \mathbf{F}_{\text{thrust}} + \mathbf{F}_{\text{gravity}} + \mathbf{F}_{\text{external}}$$

- **Thrust Force (F_{thrust}):** Generated by the propellers and acts in the body frame's z -axis:

$$\mathbf{F}_{\text{thrust}} = R_{BI} \begin{bmatrix} 0 \\ 0 \\ F_T \end{bmatrix}$$

where F_T is the total thrust, and R_{BI} is the rotation matrix from the body frame to the inertial frame.

- **Gravitational Force (F_{gravity}):** Acts in the inertial z -direction:

$$\mathbf{F}_{\text{gravity}} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}$$

- **External Forces (F_{external}):** Includes aerodynamic drag or wind disturbances.

Rotation Matrix

The rotation matrix R_{BI} is given by:

$$R_{BI} = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}$$

where c and s denote the cosine and sine of the respective angles.

Translational Dynamics in the Inertial Frame

The translational dynamics in the inertial frame are:

$$m\mathbf{a} = R_{BI} \begin{bmatrix} 0 \\ 0 \\ F_T \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}$$

Dividing by m , the acceleration becomes:

$$\mathbf{a} = R_{BI} \begin{bmatrix} 0 \\ 0 \\ \frac{F_T}{m} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix}$$

4.3 Rotational Dynamics

Angular Motion

Euler's rotational equations describe the angular motion:

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \boldsymbol{\tau}$$

where:

- $\mathbf{I}$ is the moment of inertia matrix:

$$\mathbf{I} = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix}$$

- $\boldsymbol{\omega} = [\omega_\phi, \omega_\theta, \omega_\psi]^T$ is the angular velocity vector.

Torques

Torques are generated by motor thrusts:

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix}$$

where:

- Roll torque: $\tau_\phi = b(T_2 - T_4)$
- Pitch torque: $\tau_\theta = b(T_3 - T_1)$
- Yaw torque: $\tau_\psi = d(T_1 - T_2 + T_3 - T_4)$

Here, b is the arm length, and d is the yaw control factor.

Angular Acceleration

The angular acceleration is given by:

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1} (\boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}))$$

4.4 Combined System of Equations

Translational Motion

$$\frac{d\mathbf{p}}{dt} = \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = \mathbf{a}$$

Rotational Motion

$$\frac{d\boldsymbol{\omega}}{dt} = \mathbf{I}^{-1} (\boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}))$$

Orientation Dynamics

The Euler angle rates are related to angular velocity:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

4.5 State-Space Representation

Define the state vector:

$$\mathbf{x} = \begin{bmatrix} \mathbf{p} \\ \mathbf{v} \\ \boldsymbol{\omega} \\ \boldsymbol{\phi} \end{bmatrix}$$

where $\boldsymbol{\phi} = [\phi, \theta, \psi]^T$. The combined system of equations becomes:

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} \mathbf{v} \\ \mathbf{a} \\ \dot{\boldsymbol{\omega}} \\ \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \boldsymbol{\omega} \end{bmatrix}$$

4.6 Control Inputs

1. Total thrust (F_T):

$$F_T = T_1 + T_2 + T_3 + T_4$$

2. Torques ($\tau_\phi, \tau_\theta, \tau_\psi$): Defined by individual motor thrusts as described in Section 4.2.

4.7 Simulation Results and Analysis

4.7.1 Position Over Time (X, Y, Z)

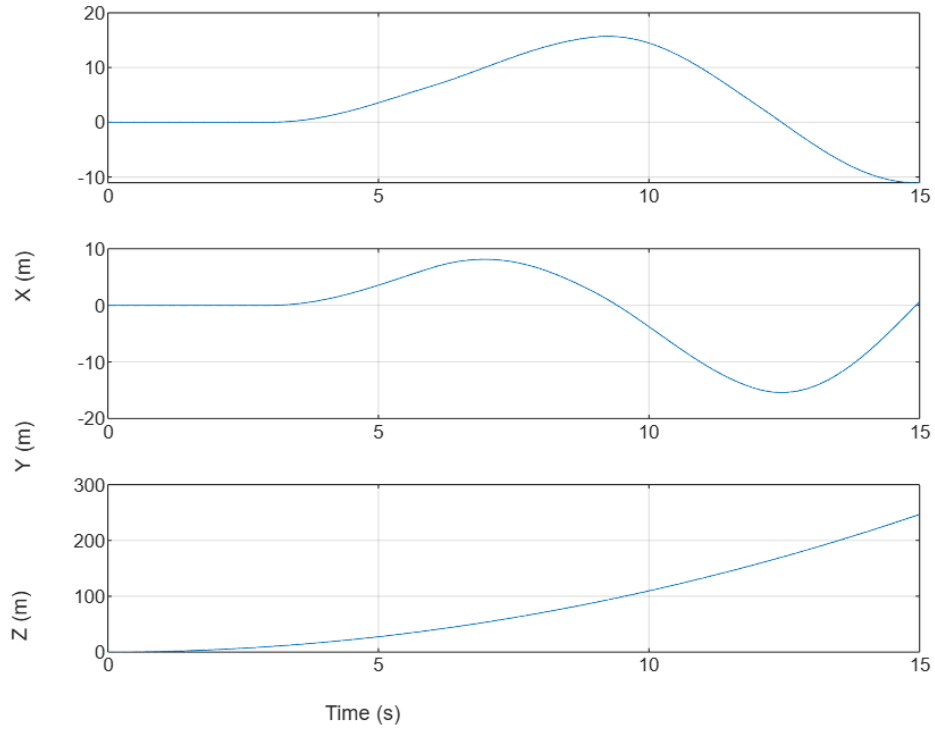


Figure 4.1: Position of the drone over time in X, Y, and Z directions.

Description:

This graph depicts the trajectory of the drone in the three axes over time: X , Y , and Z . These subplots plot the trajectory of the drone as it travels in each direction through space.

Analysis:

- The subplot for the X -axis is sinusoidal, meaning the drone is oscillating laterally over time.

- The Y-axis subplot is sinusoidal but with a phase shift compared with X-axis, as it reflects the composite nature of the trajectory of the drone in 3D space.
- The Z-axis subplot shows a steady increase, which corresponds to the drone ascending continuously during the simulation. This upward trend is consistent with the applied thrust overcoming gravity.

4.7.2 3D Trajectory

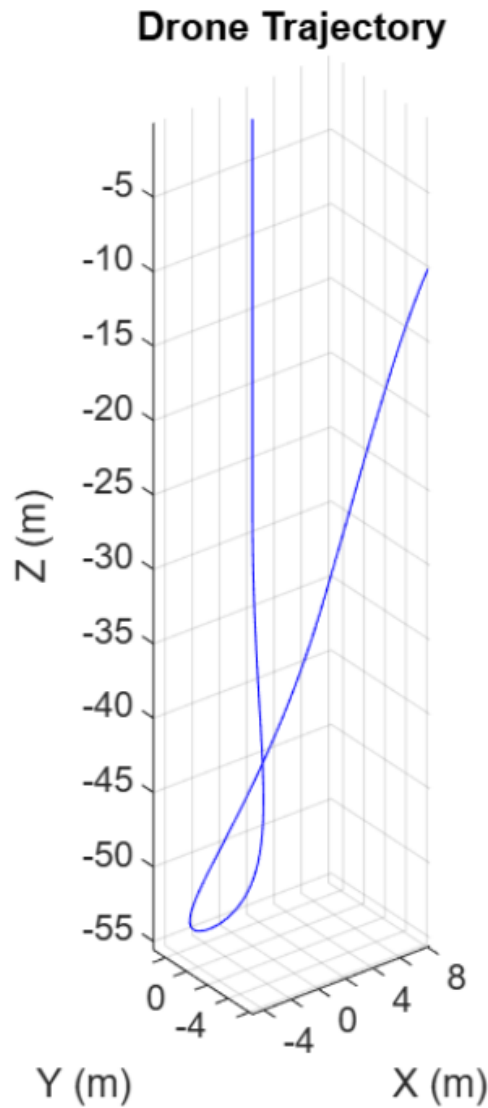


Figure 4.2: 3D Trajectory of the drone showing its path through successive waypoints.

Description:

In this 3D plot, the spatial path that the drone takes as it traverses between suc-

cessive waypoints is demonstrated. The red points show the waypoint specified and the actual flight path of the drone in terms of a blue curve.

Analysis:

- The drone, without any hitch, crosses near all waypoints in a smooth trajectory.
- Deviations from a straight line between waypoints indicate responsiveness of the drone to both control system and aerodynamic effects are such that smooth transitions between navigation targets occur.
- It has curves and changes in slope that describe dynamic adjustments in the control inputs necessary to establish stability and reach the target waypoints.

4.7.3 Cumulative Energy Consumption

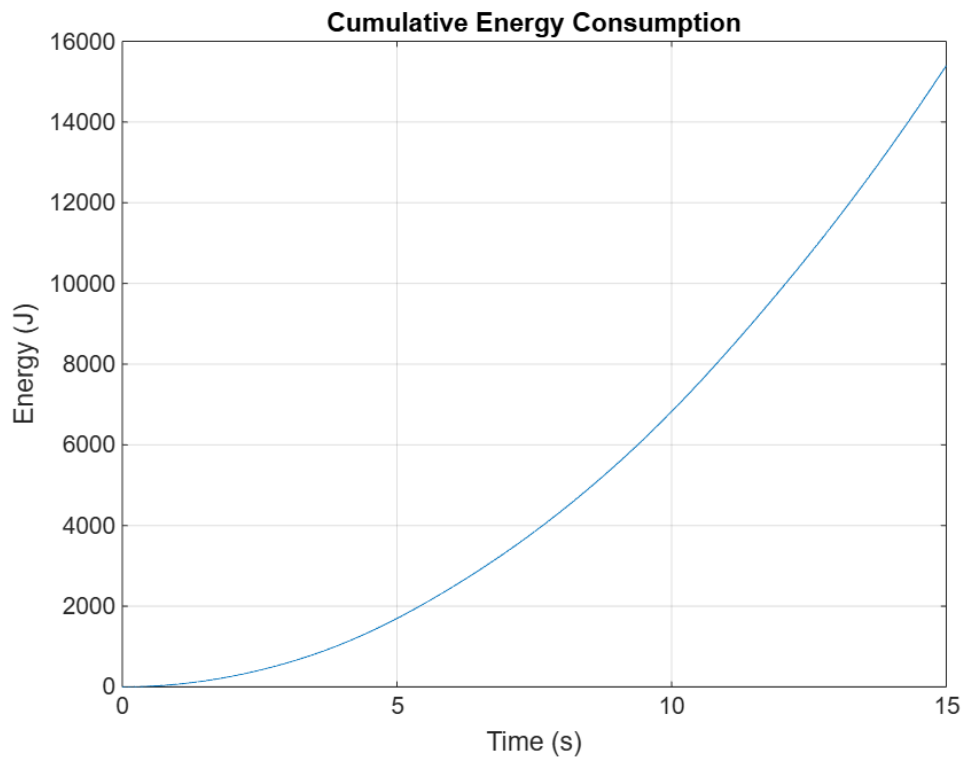


Figure 4.3: Cumulative energy consumption of the drone during the simulation.

Description:

This graph represents the cumulative energy consumption of the drone during the simulation. Energy is calculated as a function of thrust force and velocity over time.

Analysis:

- The curve of energy consumption is not linear, with a progression that increases slowly at the start and more sharply towards the end of the simulation.
- Higher demands in energy are indicated with a steeper slope due to increased thrust when ascending or attaining higher speeds, for example.
- Increasing energy consumption overall supports the fact that the drone remains constantly in climb and moves through all waypoints.

Chapter 5

Simulation

We have used various software to ensure the proper working of the drone. Here is a list of software used:

Autodesk Fusion 360: Using the Autodesk Fusion 360 software, we have designed the chassis of the drone. We have added the main frame, motors, rotors, propellers, propeller guards, camera etc. We have also kept slots for the battery and other sensors.

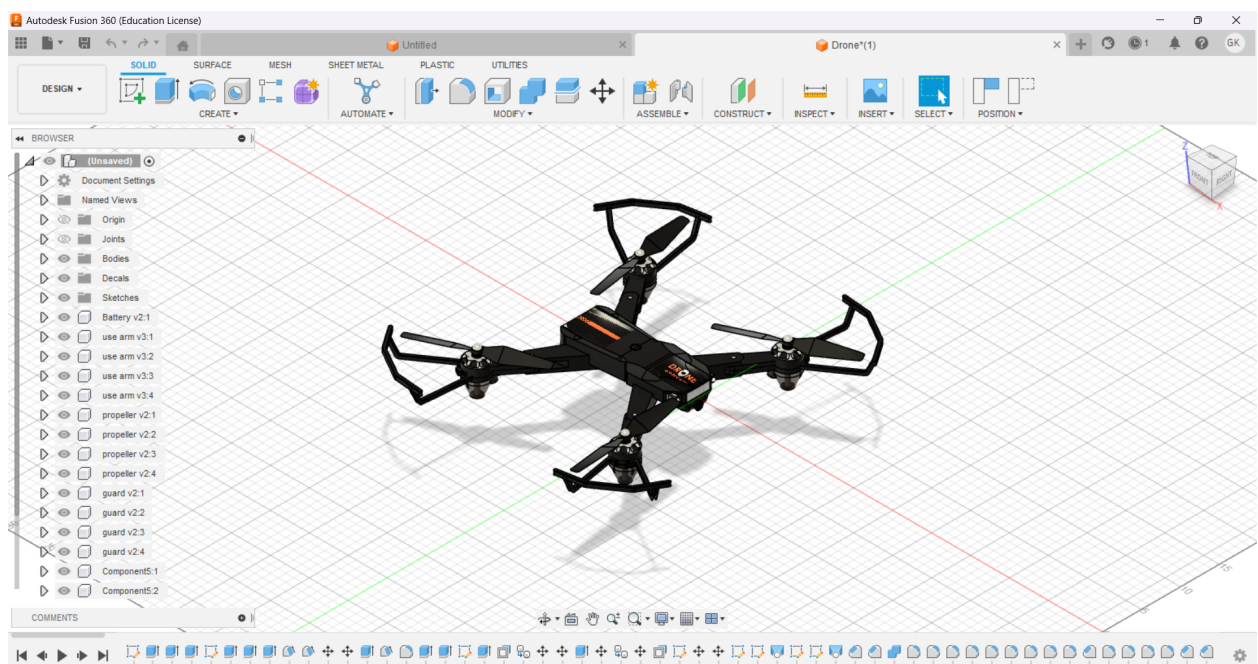


Figure 5.1: Fusion360 Drone Design Image 1

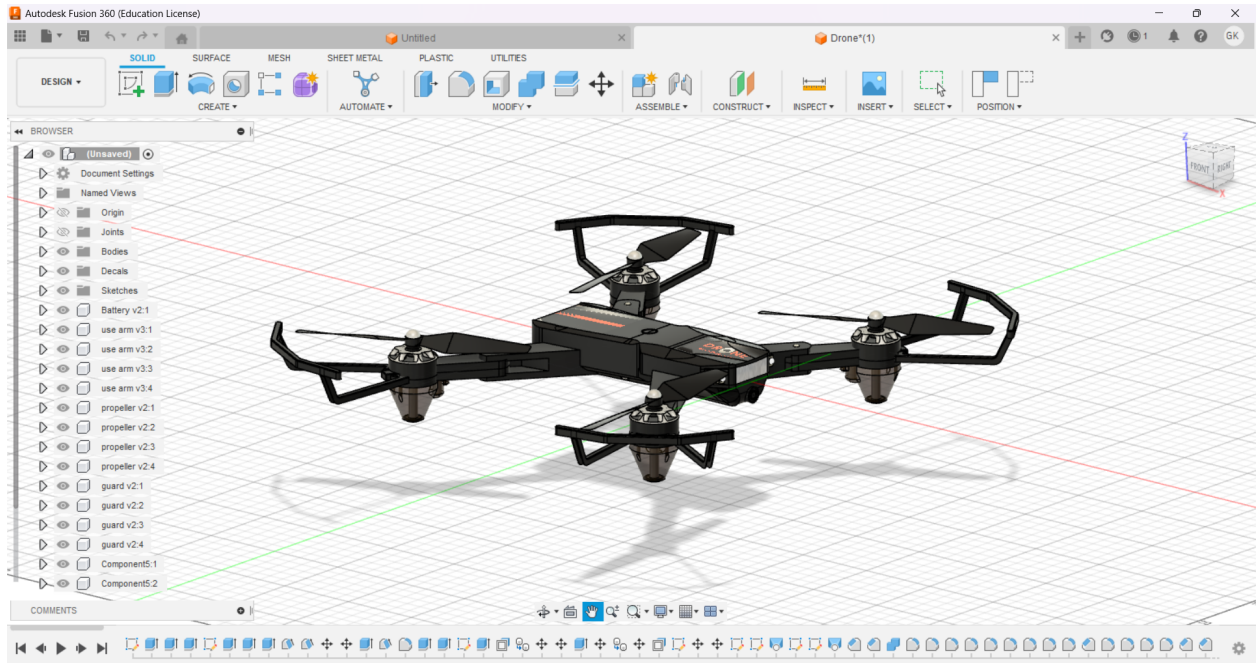


Figure 5.2: Fusion360 Drone Design Image 2

Simulink: We have used Simulink to simulate the flight pattern of the drone in various environmental conditions.

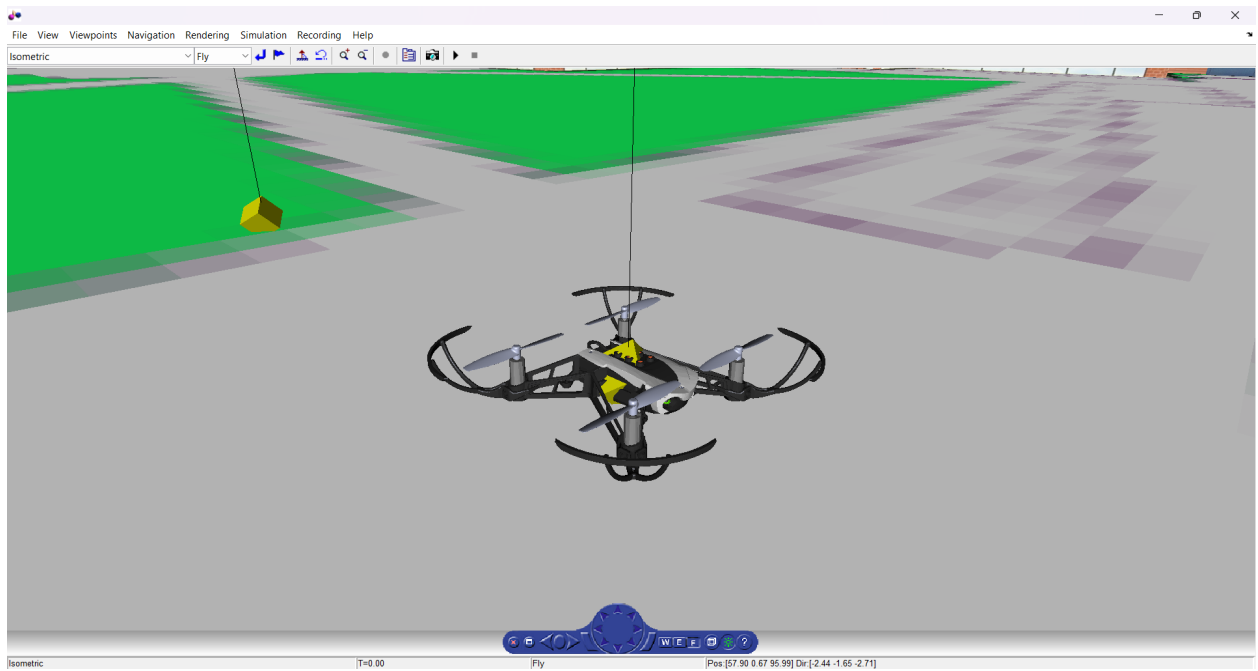


Figure 5.3: Simulink Drone Simulation Image 1

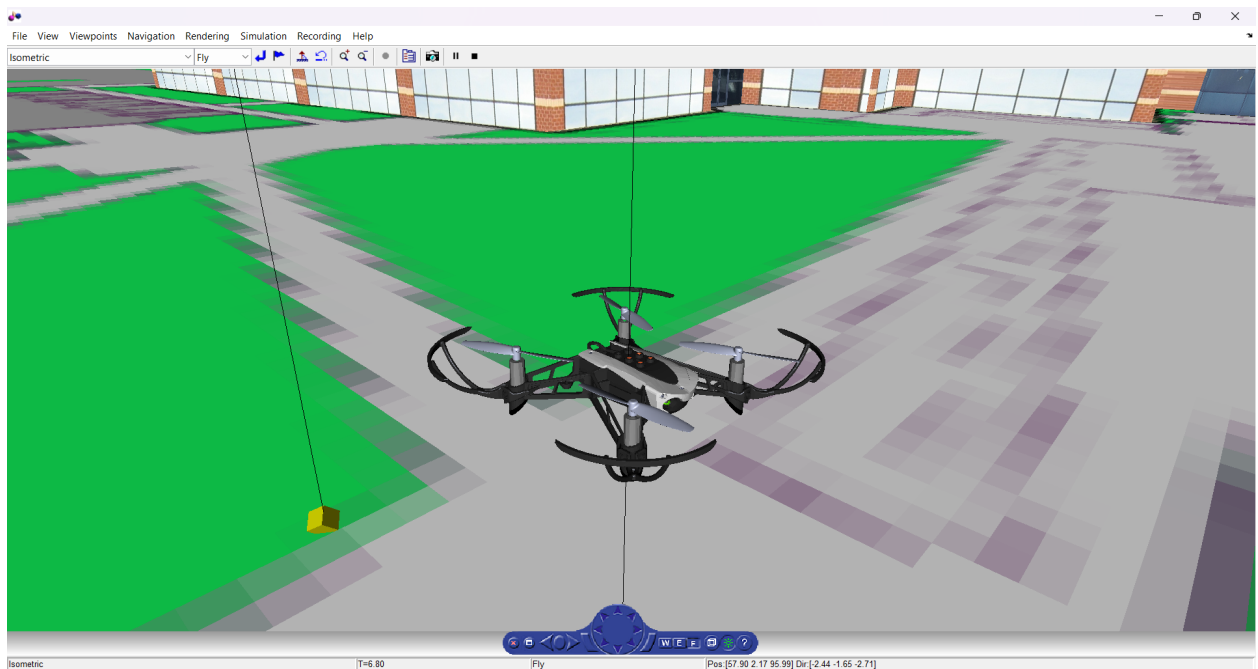


Figure 5.4: Simulink Drone Simulation Image 2

Chapter 6

Drone Images

This section displays images of the drone during setup and in the field.

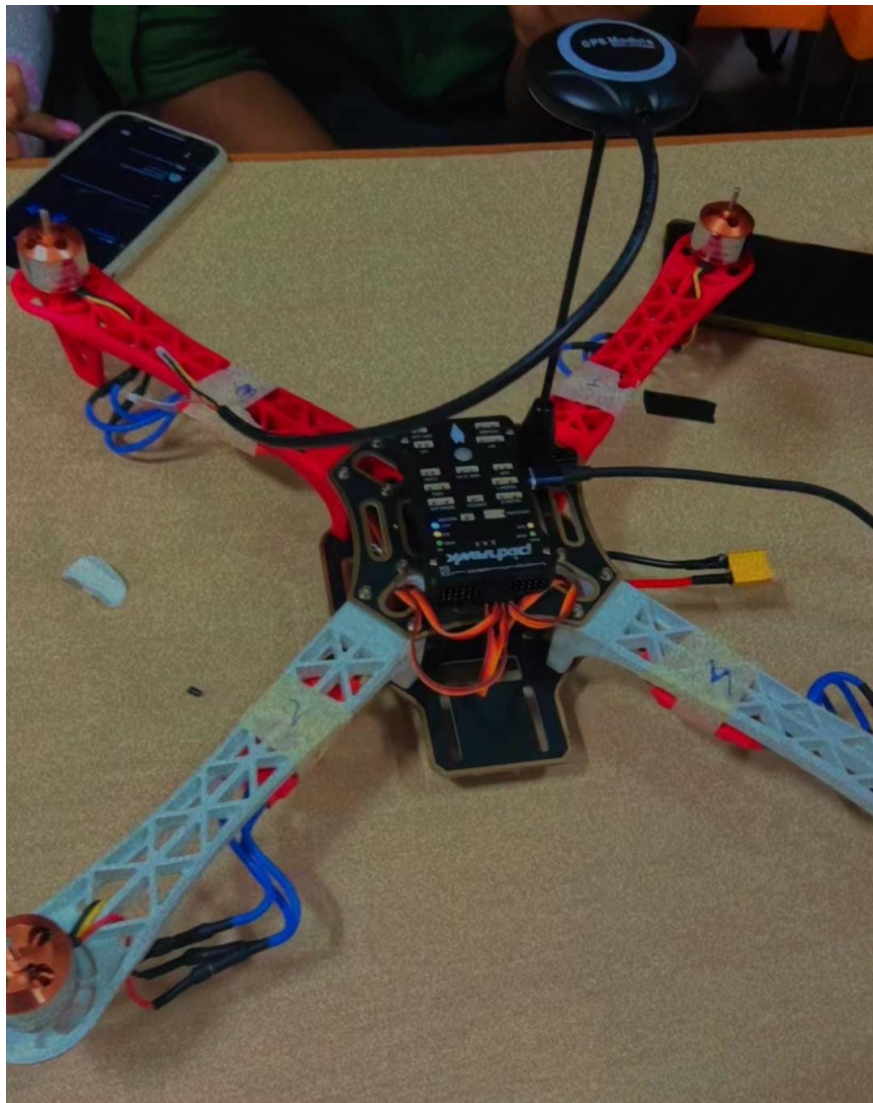


Figure 6.1: Drone - Initial Stage



Figure 6.2: Drone - Field testing

Chapter 7

Conclusion

This project has laid the groundwork for the development of an autonomous aerial system for precision crop monitoring. Through the design of a customized drone and initial testing with a smaller prototype, we have gained valuable insights into the hardware and software considerations necessary for this project.

The future development of this system will focus on integrating the 5kg drone with the gimbal-stabilized camera. Following this, the image processing algorithms for crop health assessment will be implemented on the ground station computer.

The project has the potential to revolutionize agricultural practices. By providing real-time data on crop health, this system can empower farmers to make informed decisions regarding irrigation, fertilization, and pest control. This translates to increased crop yields, improved resource management, and ultimately, a more sustainable agricultural industry.

Furthermore, the core functionalities of this system can be adapted for various applications beyond agriculture. Potential applications include environmental monitoring, search and rescue operations, and infrastructure inspections. With further development and refinement, this intelligent aerial system holds immense promise for enhancing efficiency and data acquisition across a wide range of industries.